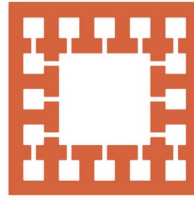


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Varendra University

Dept. of Computer Science & Engineering (CSE)

Lab 2

Course Title: **Microprocessor and Assembly Language Lab**

Course No: **CSE 3206**

Prepared By:
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Experiment No: 02

Name of the Experiment:

- i. Write an assembly language program to print a string using EMU8086.
- ii. Write an assembly language program to add two numbers using EMU8086.
- iii. Write an assembly language program to subtract two numbers using EMU8086.

```
01 .model small
02 .stack 100h
03 .code
04
05 main proc
06
07
08     exit:
09     mov ah, 4ch
10     int 21h
11     main endp
12 end main
```



Return

i. Write an assembly language program to print a string using EMU8086.

```
.model small
.stack 100h

.data
a db 'Dept. of CSE $';Variable declaration

.code

main proc

    mov ax,@data ;Initialize the data segment to code segment
    mov ds,ax

    mov ah,9
    lea dx,a ;Load Effective Address
    int 21h

    exit:
    mov ah,4ch
    int 21h
    main endp
end main
```

ii. Write an assembly language program to add two numbers using EMU8086.

```
main proc
    mov ah,1
    int 21h
    mov bl,al

    mov ah,1
    int 21h
    mov bh,al

    add bl,bh ; bl=bl+bh
    sub bl,48 ; Show output

    mov ah,2
    mov dl,10
    int 21h
    mov dl,13
    int 21h

    mov ah,2
    mov dl,bl
    int 21h
```

iii. Write an assembly language program to subtract two numbers using EMU8086.

```
mov ah,1
int 21h
mov bl,al

mov ah,1
int 21h
mov bh,al

sub bl,bh ; bl=bl+bh
add bl,48 ; Show output

mov ah,2
mov dl,10
int 21h
mov dl,13
int 21h

mov ah,2
mov dl,bl
int 21h
```